EduManager Pro

Product Requirements Document

The World's First Preschool Management System with Built-in Social Media Marketing Engine

Developed by MNTTA — Montessori & Nursery Teachers Training Academy, Bangalore, India

Target: India's 2.5 lakh preschools · One-time licence · No subscriptions · Built for mobile

Document Type	Product Requirements Document (Claude Code Command)	**Platform**	Google AppScript + HTML/CSS/JS + n8n + GoHighLevel
Target Market	India — 2.5L preschools; SEA Phase 2	**Pricing Model**	₹7,990 Early Bird · ₹14,990 Standard · ₹24,990 Pro Setup (all one-time)

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1. Executive Summary & Vision

EduManager Pro is a one-time-purchase, all-in-one preschool management and marketing platform built natively for Indian preschool owners. It is the world's first school management system to combine operational management with a complete social media marketing engine — all in one product at a single lifetime price with no recurring charges, no subscriptions, and no per-user fees ever.

1.1 Problem Statement

Indian preschool owners face three simultaneous crises that no single tool addresses today:

- Administrative overload — attendance, fees, parent communication, staff management, and compliance are managed across notebooks, WhatsApp groups, and disconnected Excel files
- Marketing paralysis — 80% of preschool owners know they need to post on Instagram and Facebook but do not know what to write, how to design posts, or when to publish
- Expensive and irrelevant software — existing tools cost ₹10,000–₹50,000/year, are built for the US/UK market, address only one function, and require recurring subscription payments that school owners deeply resent

1.2 The Solution — 19 Modules, One Product, One Price

Feature / Sub-module	Requirement
Social media engine	34 auto-personalised post templates + AI image prompts + Canva guide — no competitor has this anywhere in the world
WhatsApp-first architecture	All notifications, alerts, communications, receipts, and updates via WhatsApp — matching Indian parent behaviour
School-type adaptive	Montessori and Kindergarten modes — all templates, trackers, and reports adapt automatically to the selected school type
Daily photo sharing	One-tap classroom photo to WhatsApp for all parents in a classroom — the #1 daily task Indian parents expect
Birthday management	Auto-alert + personalised WhatsApp + social media post on every child's birthday — zero misses, maximum parent goodwill
Auto-generated Parent Handbook	Professional PDF handbook auto-generated from school profile in one click — no competitor offers this
ECA Revenue Engine	Complete post-school-hour activity management: Music, Dance, Karate, Art, Summer Camp — additional ₹1.5–₹2.5L/month revenue for a 50-student school
India-native payments	UPI QR receipts, NEFT, cash/cheque log, Tally CSV export — not ACH or Stripe
One-time price forever	₹14,990 once — replaces ₹20,000–₹75,000/year in annual subscriptions from competing tools

2. School Type Configuration

EduManager Pro supports TWO school types only. Simplified from three to reduce setup confusion and maintenance overhead. Schools that blend both methods (Integrated) select Montessori as their type and use the subject-grade fields in addition to Montessori material tracking.

Feature / Sub-module	Requirement
Montessori	Child-led work cycles · Material progression tracking (5 areas: Practical Life, Sensorial, Language, Mathematics, Cultural) · Observation-based assessment · MNTTA ECCE curriculum alignment · NEP 2020 outcome mapping · Child Portfolio PDF report · Montessori-specific social media templates and language throughout the app
Kindergarten (Traditional)	Structured teacher-led sessions · Subject timetable (English, Maths, EVS, Hindi, Art, Music, PE) · Unit plan templates · Grades-based assessment · Traditional report card · Activity-focused and festive social media templates · CBSE/ICSE-aligned terminology throughout the app
School type selection	Selected once during 5-minute onboarding · Shown as a badge throughout the app · Can be changed in Settings anytime — all existing data is preserved · Changes which templates, reports, and curriculum screens are shown by default

3. Pricing Architecture & Licence Management

3.1 Single-School Pricing Tiers

Feature / Sub-module	Requirement
Early Bird — ₹7,990 (one-time)	First 100 buyers only · All 19 modules · Lifetime access · Email support 48hr · Live counter on sales page · Creates genuine impulse-buy urgency
Standard — ₹14,990 (one-time)	Permanent price after early bird closes · All 19 modules · Lifetime access · WhatsApp support 24hr · 1 free 30-minute onboarding call · Hero tier
Pro + Setup — ₹24,990 (one-time)	All Standard features + GoHighLevel CRM configured + n8n automation workflows set up + MNTTA expert 3-hour setup session + 3 months priority support · Price anchor tier

3.2 Licence Record & Plan Upgrade Management

Feature / Sub-module	Requirement
Licence record fields	school_id · owner_name · email · phone · plan_tier · purchase_date · payment_mode · payment_reference · upgrade_history (JSON array) · support_entitlement · onboarding_call_status · school_type · active flag
Plan upgrade flow	Owner → Settings → My Plan → View current plan and benefits → Click Upgrade → See delta benefit description → Pay difference (auto-calculated) → Enter payment reference → Plan updates instantly → WhatsApp confirmation sent
No downgrade policy	One-time licence is permanently activated at the tier purchased. Communicated clearly at checkout.
7-day refund window	If fewer than 20% of features used (tracked via access log). Handled by MNTTA support team.
Licence transfer	Allowed once per licence with documentation. ₹999 transfer fee. MNTTA support team handles.

3.3 Multi-Branch / Franchise Pricing

Enterprise Plan	Branches	Price (One-Time)	Per-Branch Cost
Starter Network	2–5 branches	₹34,990	₹7,000–17,500
Growth Network	6–10 branches	₹54,990	₹5,499–9,165
Regional Chain	11–25 branches	₹99,990	₹4,000–9,090
National Franchise	26–50 branches	₹1,74,990	₹3,500–6,730
Enterprise Custom	51+ branches	Custom quote	~₹2,500/branch

3.4 Multi-Branch Feature Requirements

Feature / Sub-module	Requirement
Franchisor dashboard	Consolidated view: total enrolment across all branches · fee collection per branch · attendance % · marketing activity score (posts/month) · staff count · open leads
Branch management	Add/remove branches · set branch-specific school profile while inheriting franchise brand kit · assign branch managers with limited-role login
Brand control	Franchisor sets master logo, colours, font · branches cannot override · franchisor can batch-update all branch profiles simultaneously
Data isolation	Each branch's data is isolated from other branches · only franchisor sees all branches · zero cross-branch data leakage
Consolidated reporting	Monthly PDF: total revenue, total enrolment, occupancy % per branch, attendance trends, social media posts published

4. Module Specifications — 19 Modules

All 19 modules are included in every pricing tier. Build sequence: Phase 1 = M1, M3, M8. Phase 2 = M2, M4, M5, M6. Phase 3 = M7, M9, M10, M12, M13, M15, M18, M19. Phase 4 = M11, M14, M16, M17. Each module section specifies: sub-features, data fields, business rules, and UX behaviour.

MODULE 1 — School Identity & Configuration

Feature / Sub-module	Requirement
School Profile (22 fields)	School Name · Tagline · Full Address · City · PIN · State · Primary Phone · WhatsApp · Email · Website · Facebook · Instagram · YouTube · Google Business · Principal Name · Year Founded · Max Seats/Batch · Monthly Tuition Fee · Camp Fee · Camp Dates · Early Bird Deadline · Tour Schedule. All fields stored as Named Ranges in _config tab. Every module and all 34 social media templates auto-fill from these fields.
School type selection	Montessori or Kindergarten — selected at setup. Displayed as a badge on every screen. Changes all downstream templates, assessment tools, social media language, and report formats. Changeable in Settings with data preservation.
Brand kit	Upload school logo (PNG, transparent background) · Primary colour hex · Secondary colour hex · Font choice (Noto Sans recommended for multilingual support). All templates auto-apply brand kit.
Academic year config	Set academic year dates · Batch names (e.g. June 2026, January 2027) · Term dates · Indian public holidays pre-loaded · School-specific holidays added here
Language preference	UI language: English (default) · Kannada · Hindi · Telugu · Tamil — selected at setup. Student and parent names stored in chosen script.
School timing config	School open time · School close time · Late pickup threshold (minutes after close that triggers late pickup fee) · Late pickup fee amount per 15 minutes

MODULE 2 — Student Profiles & Enrolment

Feature / Sub-module	Requirement
Student master record	student_id (STU-YYYYMMDD-NNN) · full_name · preferred_name · dob · age (auto-calculated) · blood_group · photo · aadhaar (optional) · previous_school · batch_id · classroom_id · status (Active/Paused/Withdrawn) · deleted · deleted_at
Family profile	family_id · student_id (FK) · father_name · mother_name · guardian_name · primary_phone · whatsapp · email · occupation · home_address · emergency_contact_1 (name + phone + relationship) · emergency_contact_2 · authorised_pickup_list (JSON: name, phone, relationship, photo)

Medical record	medical_id · student_id (FK) · allergy_list (JSON) · chronic_conditions · current_medications · doctor_name · doctor_phone · hospital_preference · vaccination_record · last_physical_exam_date
Medication administration log	med_log_id · student_id (FK) · date · medicine_name · dose · time_given · given_by (staff_id FK) · parent_instruction_text · notes. Parent sends medicine with written instruction. Staff logs each administration. Parent receives WhatsApp confirmation: "[Child name] was given [medicine] [dose] at [time] today. — [Staff name]"
Sibling linking	sibling_group_id field on student record · when two students share the same family_id, sibling discount applies automatically to the younger child's fee calculation
Document checklist	birth_certificate · address_proof · parent_id · medical_cert · vaccination_card · photo (checked against uploaded files) · eConsent forms status
Student ID card	Auto-generate printable ID card: student photo, name, class, school name, emergency contact, blood group · Download PDF · Print A6 size
Enrolment record	enrolment_id · student_id · batch_id · enrolment_date · classroom_id · fee_plan_type (Monthly/Quarterly/Half-yearly/Annual) · admission_fee_paid · status

MODULE 3 — Admissions CRM & Lead Management

Feature / Sub-module	Requirement
QUICK ADD — homepage floating button	Floating "+" button always visible on homepage. Tapping opens a 4-field mini-form: Parent Name · Phone · Child Age · Source (WhatsApp/Instagram/Facebook/Walk-in/Referral/Other). Saves in 10 seconds. Full details filled later from the lead record. Business rule: this is the #1 most-used admissions feature — it must be accessible in 1 tap from anywhere in the app.
Lead pipeline stages	Enquiry → Tour Scheduled → Tour Completed → Free Trial Booked → Trial Completed → Application Submitted → Enrolled → Waitlisted → Lost. Drag-and-drop status update. Batch-centric: Indian June and January intake model (not rolling US-style).
Free Trial Day manager	Manage trial slots (max 4 per session) · Booking calendar · Confirmation WhatsApp auto-sent on booking · Post-trial follow-up reminder at 24 hours via n8n · Convert-to-enrolled one click → student profile auto-created
Source tracking & analytics	Track lead source for every enquiry · Monthly report: Instagram vs Facebook vs WhatsApp vs referral vs walk-in · Lead-to-enrolment conversion rate per source · Identify highest-ROI channel
Follow-up automation	n8n trigger: if lead status unchanged for 48 hours → WhatsApp reminder to owner. If trial completed and not enrolled in 3 days → auto-send follow-up WhatsApp template to parent (pre-built, personalised).
Batch occupancy tracker	Real-time seats available per batch per age group · Auto-alert to owner when batch is 80% full · When 5 seats remaining: urgency social media post template auto-generated in M8
GoHighLevel sync	All leads sync to GoHighLevel pipeline via n8n webhook · Two-way status update · Owner can manage follow-up from GoHighLevel if preferred

Admission letter	One-click admission confirmation letter: school letterhead + student name + batch + fee details + start date · Download PDF · WhatsApp to parent

MODULE 4 — Attendance Tracking

Feature / Sub-module	Requirement
DEFAULT: Mark All Present first	Attendance screen opens with ALL students marked PRESENT by default. Teacher only taps names to mark Absent, Late, or Half-Day. This is the 80/20 design — 95% of days, only 1-3 changes needed. Business rule: the old approach of marking each child individually is explicitly wrong for this context.
QR code check-in (optional)	Generate unique QR per student · Teacher scans on arrival · Timestamp recorded · Auto-marks present · Overrides default-present state
Morning health screening log	New section within daily attendance flow: after marking attendance, teacher records gate health check: child_name · temperature (°C or °F) · health_status (Healthy / Fever / Rash / Cough / Sent Home) · notes. If status = Sent Home: auto-WhatsApp to parent immediately: "We noticed [child name] had [health issue] today and asked you to take them home. Please rest for 48 hours. — [School Name]"
WhatsApp arrival alert	On check-in: n8n trigger sends WhatsApp to parent: "[Child name] has arrived safely at [School Name] at [time]. Have a great day!" Toggle on/off per parent preference. Major daily trust-builder.
Absence notification	If student not checked in by 9:30 AM: auto-WhatsApp to parent: "We have not seen [child name] today. Is everything okay? Please let us know. — [School Name]". Owner can disable per student if parent has pre-notified.
Pickup log (safety critical)	New log within attendance: when child is collected, staff records: pickup_person_name · relationship · time_of_collection · verified_against_authorised_list (yes/no) · collected_by_staff_id. This log is linked to the daily attendance record. Business rule: child must only be released to a person on the authorised pickup list. System shows authorised list photo when logging pickup.
Late pickup tracking	If child not picked up by school close time + late pickup threshold (configured in M1): system auto-calculates late pickup fee · shows on next fee collection screen · WhatsApp sent to parent: "[Child name] was picked up at [time], [X] minutes after school close. A late pickup fee of ₹[amount] has been added."
Teacher login attendance	Teacher logs in with limited credentials → sees ONLY their classroom → marks attendance for their own students only. Owner sees all classrooms. Teacher cannot access fee, CRM, or other modules.
Staff attendance	Separate staff daily check-in · Leave request and approval workflow · Leave balance (CL/PL/SL per Indian labour norms) · Monthly staff attendance summary
Substitute teacher assignment	When a teacher marks themselves absent OR owner marks a teacher absent: system shows "Classroom [X] is uncovered" alert · Owner assigns substitute from available staff · System updates ratio dashboard · WhatsApp sent to parents of affected classroom: "Ms [Name] is unwell today. [Substitute name] will be with [Classroom Name]. — [School Name]"
Monthly attendance report	Per student: days present/absent/late/health-sent-home · Attendance % · Annual register PDF · Format aligned to school inspection requirements

MODULE 5 — Fee Management

Feature / Sub-module	Requirement
Fee plan types (enhanced)	At enrolment, select fee payment plan per student: Monthly (default) · Quarterly (April, July, October, January) · Half-yearly (April, October) · Annual (April). System calculates due dates automatically for each plan. A student on Quarterly plan receives reminders only in April, July, October, January — never in between.
Fee structure setup	Configure fee heads: Tuition · Admission (one-time) · Transport · Activity · Material · Late Pickup · Any custom head. Set amount per head per plan type. Sibling discount auto-applied.
UPI receipt generator	Mark payment received · Record mode (UPI/NEFT/IMPS/Cash/Cheque) + reference · Auto-generate formatted receipt: school name · GSTIN (if applicable) · student name · fee heads · amount in words (Indian legal format) · payment reference · school stamp field · Download PDF · WhatsApp to parent immediately
Fee reminder system	AppScript trigger on fee due date (based on plan type): WhatsApp reminder to parent · Second reminder 5 days later if unpaid · Escalation to owner after 10 days: "₹[amount] outstanding for [student name] — [days] overdue"
Defaulter dashboard	Outstanding fees by student · filter by batch/classroom · total outstanding amount · months overdue · one-tap WhatsApp to individual parent · bulk WhatsApp to all defaulters
Tally CSV export	Fee collection data exportable as Tally-compatible CSV monthly · eliminates manual double-entry for accountant
Annual fee report	Fee collected vs expected per month · YTD collection · Revenue chart · Outstanding total · Export PDF for CA/accountant

MODULE 6 — Parent Communication Hub

Feature / Sub-module	Requirement
Daily classroom photo/video sharing (MUST HAVE)	Teacher or owner opens app → "Share Today's Photos" → selects classroom → takes photo with camera OR picks from gallery (max 8 images per send) → auto-resized to WhatsApp-optimal dimensions → preview shown → selects recipients: All Parents in Class / Individual Parents · One tap to send via WhatsApp broadcast template. Business rule: this is the #1 daily task that parents love and that drives daily app engagement. Must be accessible from the home screen in 2 taps.
WhatsApp broadcast generator	Pre-written templates: Holiday notice · Exam schedule · Event invite · Fee reminder · School closure · Parent meeting · Emergency · Weekly update. All auto-filled from School Profile. Select recipients: All parents / Specific batch / Individual family.
Circular generator	Create circulars with school letterhead · rich text editor · auto-insert school name/date/logo · Download PDF · WhatsApp to all parents or specific batch
eConsent forms	Field trip permission · Photo/video consent · Medical treatment consent · Event participation. Digital acknowledgement via WhatsApp reply. Status tracked per family.

Emergency alert	One-tap broadcast to ALL parents with highest priority. Includes school name, date/time, message. Non-dismissable delivery tracking.
Communication log	Full history of all broadcasts, circulars, and individual messages per family. Searchable by date/type/family.
Monthly newsletter	Template: principal message · events this month · child spotlight · upcoming dates. Download PDF · WhatsApp / Email.

MODULE 7 — PTM (Parent-Teacher Meeting) Manager

Feature / Sub-module	Requirement
PTM schedule builder	Set PTM date · time slot duration (15/20/30 min) · auto-generate slots for entire school · assign families by batch / alphabetical / manual
PTM invite	Auto-generate personalised invite per family: date, time slot, teacher name · WhatsApp to each parent · Google Calendar link
Pre-meeting child summary	Auto-generate per-child PTM summary: attendance % · fee status · material progress (Montessori) or subject grades (KG) · observation highlights · discussion points. Teacher reviews and personalises before the meeting.
Meeting notes & action tracker	Record discussion points, parent concerns, decisions, action items during PTM · All actions tagged to student profile · Action dashboard: open/closed/overdue · WhatsApp reminder on action due date
PTM feedback form	Post-meeting WhatsApp to parent: 3-question anonymous feedback. Results aggregated for management review.
PTM history	Full PTM history per child: date · teacher · summary · action items · outcomes. Accessible in student profile forever.

MODULE 8 — Social Media Marketing Engine

World-first differentiator — no school management software anywhere in the world includes a marketing engine. This module alone justifies the purchase for most Indian preschool owners who are struggling with consistent, quality social media content.

Feature / Sub-module	Requirement
34 auto-personalised templates	6 Summer Camp · 6 Admissions Open · 6 Parent Testimonials · 6 Safety PSA · 4 Free Trial · 36-post Content Calendar (April–June). School Profile fields auto-fill all [bracket] placeholders via SUBSTITUTE formulas. Montessori and KG schools see school-type-appropriate language automatically.
Birthday post template	Separate from the 34 templates: a personalised birthday social media post for each child. Auto-filled with child's first name, age, classroom, school name. One tap to post when M11 birthday alert fires. Parents screenshot and share — free organic marketing.
AI image prompt library	One detailed AI image prompt per template. Each prompt specifies: Indian children, Indian settings, Indian cultural elements (terracotta, marigolds, brass,

	dhurrie mats). Paste into Canva Magic Media, DALL-E 3, or Midjourney for professional culturally appropriate images in 30 seconds.
Content calendar (April–June 2026)	36-post calendar with: date · platform · post type · template caption (with [brackets]) · auto-personalised caption (SUBSTITUTE formula) · hashtags · CTA · Canva size · AI image prompt · status dropdown (Planned/Designing/Scheduled/Published).
Canva guide (in-app)	Brand kit setup instructions · 10 post sizes with specifications · AI prompt formula (7-part framework) · optimal posting times for Indian parents · hashtag strategy (4 tiers) · monthly posting checklist
Hashtag generator	Input post topic → generate 30 relevant hashtags across broad/city/topic/branded tiers · auto-includes school city from profile · one-tap copy
Annual content calendar (V2)	Full 12-month calendar mapped to Indian school year and festivals. V1 ships April–June 2026 only. Annual calendar is Phase 2 feature.
Testimonial collection system	Per-testimonial WhatsApp collection scripts · consent tracking · 6 testimonial post templates · video testimonial filming guide

MODULE 9 — Staff Management

Feature / Sub-module	Requirement
Staff profile	staff_id (STF-NNN) · full_name · role (Owner/Head Educator/Assistant Educator/Support Staff/ECA Trainer) · qualification · mntta_cert_number · mntta_cert_expiry · bgv_status · bgv_document_ref · aadhaar_optional · emergency_contact · join_date · status
MNTTA certification tracker	Cert expiry date stored · AppScript alert 60 days before expiry: WhatsApp to owner "Ms Anita's MNTTA certification expires on [date]. Time to renew!" · MNTTA Certified badge on staff profile · Unique to EduManager Pro — no competitor has this
Teacher profile (extended)	For role = Teacher or Assistant: classroom_assignment (primary) · secondary_classroom · age_groups_qualified · school_type_certified · teacher_login_email (for limited-role login). One teacher = one primary classroom as Head. Can be Assistant in multiple.
Substitute teacher assignment	When teacher absent: system shows uncovered classroom alert · Owner assigns substitute from available staff → ratio dashboard updates → parent notification WhatsApp auto-generated (see M4).
Staff attendance & leave	Daily check-in/check-out · leave request and approval · CL/PL/SL balance · monthly summary
Salary register	Monthly calculation: basic + allowances – deductions · PF/ESI-aware fields · mark paid · generate salary slip PDF · salary history
Appraisal system	Annual appraisal form: punctuality, parent communication, classroom management, curriculum delivery, professional development. Historical records.
Offer & appointment letters	Auto-generate with school letterhead · download PDF · WhatsApp to staff

MODULE 10 — Curriculum, Learning & Assessment

Feature / Sub-module	Requirement
QUICK OBSERVATION — 15-second entry (MUST HAVE)	Default entry mode: tap child name → select material from dropdown (pre-loaded by school type) → select status (Introduced / Practising / Mastered) → optional 1-line free text → SAVE. Business rule: the full structured observation form exists for detailed notes but MUST NOT be the default. Observing during a Montessori work period requires silent, non-disruptive 15-second logging.
Montessori material progression tracker	Five curriculum areas: Practical Life · Sensorial (Pink Tower, Red Rods, Broad Stair, Cylinder Blocks, Colour Boxes...) · Language (Sandpaper Letters, Moveable Alphabet, Phonics...) · Mathematics (Number Rods, Sandpaper Numbers, Golden Beads...) · Cultural. Per-child material progress map. Status: Not Introduced / Introduced / Practising / Mastered. Only shown for Montessori school type.
KG lesson plan template	Weekly lesson plan: subject, objective, activity, materials, assessment method · NEP 2020 competency linkage · Print-ready PDF. Only shown for KG school type.
NEP 2020 alignment	Every learning activity linkable to a NEP 2020 Early Childhood Competency. Compliance report. Unique differentiator for school inspections.
Child observation log (full)	Teacher records observations per child: concentration span · social interaction · independence level · emotional state · free text. Searchable by date/child/area.
Progress report / Portfolio	Montessori: child portfolio PDF compiled from observations + material progress + teacher comments. KG: subject-wise grades + teacher comments + attendance. Download PDF · WhatsApp to parent at PTM.
Monthly activity planner	Monthly calendar for classroom. Pre-loaded with Indian festival activities, seasonal themes, Montessori work cycle plans. Editable.

MODULE 11 — Events, Activities & School Calendar

Feature / Sub-module	Requirement
Monthly calendar view (GUI)	Grid of 30/31 days. Colour-coded day cells: Navy dot = school events · Purple = PTM slots · Orange = ECA sessions · Red = holidays · Gold = fee due dates · Green = admissions events. Tap any day → day panel showing all items. Navigate month forward/back. Export as PDF for parents.
Weekly calendar view (GUI)	7-column grid Mon–Sun, 7:00 AM–7:00 PM in 30-min increments. Shows: regular school hours (shaded) · ECA batches (orange blocks, trainer name) · PTM slots (purple blocks, parent name) · staff leave (grey). Classroom filter. Doubles as timetable manager for ECA scheduling.
Add event from calendar	Tap any date/slot → Add Event → type (School Event/PTM/ECA/Holiday/Other) → fill details → save. Shows immediately on calendar. Optional WhatsApp notification.
Birthday management (MUST HAVE)	System checks every morning at 7:00 AM: are there any birthdays today or tomorrow? If yes: dashboard banner alert with child photo, name, classroom. Auto-generate personalised WhatsApp message to parent: "Happy Birthday to our wonderful [child name]! [School Name] wishes [him/her] a magical [age]th

	birthday!" One-tap birthday social media post (from M8 birthday template). Zero missed birthdays — ever.
Annual school calendar	Pre-loaded Indian public holidays + school holidays. Add custom events. Share as PDF or WhatsApp broadcast.
Event planner	Annual Day / Sports Day / Graduation / Grandparents Day / Field Trip / PTM · checklist template · budget tracker · venue details
Graduation ceremony planner	Student list with roles · certificate generator (auto-fill name, school, year) · parent invite letter · social media post linked to M8
Summer camp planner	This is the marketing/content planner — operational Summer Camp management lives in M19 ECA module. Here: weekly theme calendar · daily activity plan · parent communication templates
Festival activity guide	Pre-built guides: Diwali · Holi · Dussehra · Ganesh Chaturthi · Pongal/Sankranti · Independence Day · Republic Day · Teachers Day · Children's Day · Eid · Christmas
Google Calendar sync	All events, PTMs, ECA batches sync to owner Google Calendar. Parent PTM invites include Google Calendar link.

MODULE 12 — Transport Management

No competitor including Procare has a transport module. 60%+ of Indian preschool children use school transport. This module is a significant differentiator.

Feature / Sub-module	Requirement
Vehicle register	vehicle_id · number · make/model · capacity · insurance_expiry · fitness_cert_expiry · driver_id · status. Alert 30 days before document expiry.
Driver register	driver_id · name · licence_number · licence_expiry · bgv_status · police_clearance_ref · phone · emergency_contact · route_id
Route planner	Named routes with stops in sequence · landmark descriptions · estimated arrival time per stop · assigned vehicle and driver
Student bus assignment	Assign each student to route and stop · view all students on each route · print daily pickup list per vehicle
Transport fee	Separate from tuition · monthly collection · receipt generation · outstanding tracker · exemption flag
WhatsApp route updates	Pre-built templates: "Bus departed [School] and will reach [Stop] at approx. [Time]" · "Bus delayed [N] minutes due to traffic" · one-tap send
Document expiry dashboard	All vehicle and driver documents: Green (valid) · Amber (expiring 30 days) · Red (expired). WhatsApp alert to owner on amber status.

MODULE 13 — Safety & Compliance

Feature / Sub-module	Requirement
Visitor log	visitor_id · name · purpose · contact · entry_time · exit_time · person_met · ID_verified. Daily visitor report.
POCSO compliance module	Staff awareness checklist · staff acknowledgement form (stored digitally) · mandatory reporting SOP · body safety policy template (for Parent Handbook) · safe environment audit checklist
Fire safety log	Termly drill schedule · date of last drill · evacuation time · issues noted · extinguisher inspection log (monthly) · next service date
CCTV audit log	Coverage confirmation checklist · last recording review date · storage duration confirmed · malfunction log
Accident / incident report	Structured report: date/time · child involved · description · action taken · parent notified time · medical treatment · follow-up required. Printable PDF.
Medical emergency SOP	Step-by-step emergency procedure · emergency contacts (hospital, ambulance, poison control) · pre-filled with nearest hospitals from school city in profile
Safety PSA post templates	6 India-specific posts (in M8): School Safety · Monsoon Water Safety · Road Safety · Diwali Fire Safety · Body Safety (POCSO) · Digital Screen Safety
Authorised pickup enforcement	Photo + name of every authorised pickup adult per child (from M2 family profile) · staff confirms identity before child release · pickup time and person logged in M4 pickup log

MODULE 14 — Reports & Analytics Dashboard

Feature / Sub-module	Requirement
Homepage executive dashboard	TODAY: attendance % (with one-tap to mark) · fees collected this month vs target · next event this week · posts published this month · staff on leave today · any birthdays today or tomorrow (birthday alert). FOLLOW UP: leads not contacted 48hrs · students with outstanding fees · documents expiring in 30 days
Enrolment analytics	Current enrolment by batch/age group · occupancy % · monthly trend · batch capacity utilisation · YoY comparison
Fee analytics	Collected vs expected · outstanding amount · defaulter count · revenue by fee head · annual revenue chart · quarterly/annual plan tracking
Attendance analytics	School-wide % per day/week/month · students below 75% · health screening summary (sent home count per month) · staff attendance summary
ECA revenue analytics	ECA collected vs expected · per-activity revenue · trainer cost vs revenue · net ECA profit monthly · occupancy per batch
Marketing analytics	Posts published vs target · enquiries by source · lead-to-enrolment conversion rate · free trial conversion · CRM pipeline summary
Annual school report	Year-end report: total enrolment, fee collected, average attendance, staff count, events conducted, social media posts, safety drills, ECA programmes. Download PDF.

Custom report builder	Select metrics → date range → batch/class filter → generate → download PDF or CSV

MODULE 15 — Parent Handbook Generator

Auto-generates a complete professionally formatted Parent Handbook PDF from School Profile data in one click. No competitor offers this. Every new parent receives a personalised handbook on admission day.

Feature / Sub-module	Requirement
Auto-generated content	School name, logo, tagline, address, contact · Principal welcome message · School philosophy (Montessori or KG, adapted to school type) · MNTTA certification note
Academic information	Academic year calendar · daily routine schedule · curriculum overview (school-type adapted) · assessment and reporting approach
Policy sections	Attendance policy · punctuality · holiday procedure · medication administration policy (links to M2 medication log) · allergy management · school closure communication
Safety policies	Gate safety procedure · authorised pickup rules · visitor policy · CCTV notice · emergency procedure · POCSO body safety note for parents
Fee section	Fee structure and heads · payment plan type (monthly/quarterly/annual) · payment modes (UPI, NEFT, cash, cheque) · late payment policy · refund policy
Transport section	Available routes · drop-off and pickup times · driver contact · route update WhatsApp group instructions
Parent involvement	PTM schedule · parent volunteer opportunities · classroom observation policy · communication channels
FAQ section	20 most common new-parent questions — answers adapted to school type. Owner can add custom Q&As.
Download and share	Download professionally formatted PDF with school branding · WhatsApp to parent on admission day · re-generate anytime after profile changes

MODULE 16 — Staff CPD & Training Log (Simplified)

Simplified from original spec. V1 scope: staff CPD log + MNTTA cert tracking. Full MNTTA trainee cohort management is an MNTTA internal product, not a preschool owner tool, and is removed from this PRD.

Feature / Sub-module	Requirement
MNTTA cert expiry tracker	Staff MNTTA/ECCE certification number + expiry date · 60-day pre-expiry WhatsApp alert to owner · MNTTA Certified badge on staff profile

CPD training log	Record training attended per staff member: workshop name · date · provider · certificate upload · CPD hours accumulated · type (External/Internal/MNTTA)
Internal training calendar	Schedule in-house training sessions · attendance register · notes
Training summary report	Annual CPD hours per staff member · MNTTA cert status across all staff · training budget vs actual

MODULE 17 — Integrations & Automation

Feature / Sub-module	Requirement
GoHighLevel CRM sync	All admissions leads auto-sync to GoHighLevel pipeline via n8n webhook · two-way status update · Pro Setup tier includes full GHL configuration
n8n automation workflows (pre-built)	(1) Attendance check-in → WhatsApp parent · (2) Fee due → reminder · (3) Lead stale 48hr → owner nudge · (4) Trial completed → follow-up · (5) Birthday tomorrow → owner alert + WhatsApp template ready · (6) Health screening sent-home → parent alert · (7) Staff absent → substitute alert · (8) Batch 5 seats remaining → urgency post template auto-generated in M8
WhatsApp Business API	WABA-compliant templates · message delivery receipts · opt-in/opt-out management · high-volume broadcast support
Google Calendar sync	School events, PTM appointments, ECA batches → Google Calendar · parents receive calendar link in invitations
Tally CSV export	Fee collection monthly export · Tally-compatible format · eliminates double-entry for accountant
Inbound webhooks	Facebook Lead Form → CRM quick-add · Google Form → student enrolment · New enquiry → auto-acknowledgement WhatsApp to parent

MODULE 18 — Classroom & Teacher Management

Feature / Sub-module	Requirement
Create classroom	classroom_id (CLS-NNN) · name · type (Montessori Primary/Casa/KG Nursery/LKG/UKG) · capacity · head_teacher_id · asst_teacher_id · school_type · status. Classroom cards show: name, type badge, teacher name, live occupancy bar (colour-coded: Green <80% / Amber 80-95% / Red over capacity).
Assign teacher to classroom	From classroom screen: select classroom → assign head teacher (dropdown of staff with role=Teacher) → assign assistant (optional) → save. One teacher = one primary classroom as head. Historical assignment log maintained. Changing teacher records previous assignment with date.
Assign child to classroom	Three modes: (1) At enrolment — classroom selected from rooms matching child age group. (2) Bulk assignment — select multiple enrolled students → assign all to a classroom. (3) Individual transfer — move child between classrooms with reason and date. Classroom occupancy auto-updates. Transfer history per child.

Classroom roster view	List of all enrolled students in a classroom · parent WhatsApp numbers · attendance status today · ECA activities enrolled in. Quick actions: Mark Attendance / WhatsApp Parents / View Portfolio.
Staff-to-child ratio monitoring	After every assignment: check ratio against limits (Montessori 1:8, KG 1:12). Alert if exceeded. Ratio dashboard on homepage for all classrooms.
Classroom-level attendance	Teacher (limited login) opens app → sees their classroom only → marks attendance using "All Present" default → adjusts exceptions → logs health screening. Business rule: teacher cannot see other classrooms, fees, CRM, or management reports.

MODULE 19 — Extra-Curricular Activities (ECA) Revenue Engine

REVENUE IMPACT: *Post-school-hour activities represent ₹2,000–₹5,000/child/month in additional revenue. A 50-student school with 3 ECA programmes generates ₹1.5–₹2.5L/month additional. This module manages every aspect of ECA delivery.*

19.1 Activity & Batch Setup

Feature / Sub-module	Requirement
Create ECA activity	eca_id · name (e.g. Bharatanatyam Dance, Keyboard Music, Karate, Watercolour Art, Yoga, Abacus Maths) · category (Performing Arts / Martial Arts / Visual Arts / Academic / Sports / Wellness) · description · suitable_age_groups · status
Create ECA batch	batch_id · eca_id (FK) · trainer_id (FK) · classroom_id (FK — room shared after school hours) · batch_name · days_of_week (multi-select) · time_slot_start · time_slot_end · start_date · end_date · max_students · fee_per_month · status. Conflict detection: alert if trainer or classroom double-booked at same time.

19.2 ECA Trainer Management

Feature / Sub-module	Requirement
External trainer profile	trainer_id · name · specialisation_list · phone · whatsapp · email · contract_type (Per session / Monthly / Revenue share %) · rate · bgv_status · active_batches. Separate from regular school staff.
Trainer payment tracker	Sessions delivered × per-session rate · or monthly retainer · amount paid · outstanding balance · payment history. Separate from staff salary module.
Trainer timetable	Visual weekly timetable per trainer. Conflict detection at batch creation.

19.3 ECA Enrolment

Feature / Sub-module	Requirement
Enrol student	From student profile or batch screen: select student → select ECA → select batch → check age eligibility + capacity + timing conflict → create enrolment. enrolment_id · batch_id · student_id · enrolment_date · status (Active/Paused/Withdrawn).

Enrolment confirmation WhatsApp	Auto-sent on enrolment: "[Child name] is enrolled in [Activity] at [School]. Batch: [days/time]. First session: [date]. Monthly fee: ₹[amount]. See you there!"
Batch roster	Enrolled students + parent WhatsApp numbers. One-tap batch WhatsApp: "Tomorrow's [Activity] session is confirmed. See you at [time]!" Separate ECA attendance per session.

19.4 ECA Fee Management

Feature / Sub-module	Requirement
Separate ECA fees	ECA fees are separate from tuition. Each batch has its own monthly fee. Student enrolled in 2 ECA activities has 2 separate ECA fee records. Optional combined receipt mode available.
ECA fee collection	Per batch: list enrolled students → payment status → mark paid (UPI/cash/cheque) → auto-generate ECA receipt PDF → WhatsApp to parent immediately
ECA fee reminder	AppScript: WhatsApp reminder on fee due date per batch. Separate from tuition reminder.
ECA defaulter tracker	Outstanding ECA fees dashboard. Owner can pause enrolment for 2+ months unpaid with auto-WhatsApp notice to parent.

19.5 ECA Revenue Dashboard

Feature / Sub-module	Requirement
Monthly ECA revenue card	Homepage: "ECA Revenue This Month" · collected vs expected · per-activity breakdown · trainer cost · net ECA profit
Summer Camp as ECA programme	Special time-bound ECA: camp name · dates · daily time · age group · capacity · fee · weekly theme planner (pre-loaded: Around the World/Little Scientists/Arts & Culture/Nature Warriors) · daily activity plan · camp certificate generator (PDF, WhatsApp on final day). Marketing templates for Summer Camp live in M8.
ECA occupancy report	Capacity vs enrolment · occupancy % · waitlist count · month-on-month trend · low-enrolment activities flagged
Combined revenue view	Total school revenue: Tuition + ECA + Transport. Breakdown chart. YTD comparison. Projected annual ECA revenue.

5. Data Architecture — Google Sheets Schema

One workbook per school. 26 tabs. All tabs hidden from sheet tab bar (sheet.hideSheet()). All data access via AppScript DAL only — never direct Sheet access by any user. Named ranges for all _config fields. Soft deletes only — never sheet.deleteRow().

Tab name	Type	Key columns	Protected
_config	Config	key · value (22 rows, all Named Ranges — SchoolName, SchoolPhone, CampDates etc.)	Full
_audit_log	System	timestamp · user_role · action · table · row_id · field · old_val · new_val	Append-only
_deleted	Recovery	original_table · original_row_id · full_row_json · deleted_by · deleted_at · restored	Append-only
_backup_log	System	backup_id · timestamp · type (daily/weekly) · rows · status · destination	Append-only
_licence	System	school_id · plan_tier · purchase_date · payment_ref · upgrade_history_json · support · active	Full
students	Data	student_id · name · dob · age · batch_id · classroom_id · fee_plan_type · status · deleted · deleted_at	Full
families	Data	family_id · student_id · parent1 · parent2 · guardian · primary_phone · whatsapp · email · pickup_list_json	Full
medical	Data	medical_id · student_id · allergies_json · conditions · medications · doctor · hospital · vacc_record	Full
med_log	Data	log_id · student_id · date · medicine · dose · time_given · given_by_staff_id · parent_instruction	Full
attendance	Data	att_id · student_id · date · status · check_in_time · health_status · health_temp · marked_by	Full
pickup_log	Data	pickup_id · student_id · att_id · collector_name · relationship · time · verified_against_list · staff_id	Full
health_screen	Data	screen_id · student_id · date · temperature · status · sent_home · notes · recorded_by	Full
fees	Data	fee_id · student_id · month · fee_head · amount · plan_type · mode · upi_ref · paid_at · receipt_no	Full
leads	Data	lead_id · parent_name · child_age · source · status · last_contact · batch_target · notes	Full
staff	Data	staff_id · name · role · classroom_id · mntta_cert · cert_expiry · bgv_status · join_date · status	Full
classrooms	Data	classroom_id · name · type · capacity · head_teacher_id · asst_teacher_id · school_type · status	Full
classroom_stu	Data	cs_id · classroom_id · student_id · assigned_date · transferred_from · transfer_reason	Full

ptm	Data	ptm_id · student_id · staff_id · date · slot · notes · actions_json · status	Full
transport	Data	transport_id · vehicle_no · driver_id · route · student_ids_csv · fee	Full
safety_log	Data	log_id · type (visitor/drill/incident) · date · details · action_taken · resolved	Full
curriculum	Data	curr_id · student_id · material_name · area · status · introduced · mastered · notes	Full
communications	Data	comm_id · type · recipient_ids_csv · channel · sent_at · content_ref · status	Full
events	Data	event_id · name · date · type · checklist_ref · budget · status	Full
eca_activities	Data	eca_id · name · category · age_group · description · status	Full
eca_batches	Data	batch_id · eca_id · trainer_id · classroom_id · days · time · capacity · fee · start · end · status	Full
eca_enrolments	Data	enrol_id · batch_id · student_id · enrol_date · status	Full
eca_fees	Data	fee_id · enrol_id · month · amount · mode · reference · paid_at · receipt_no	Full
eca_trainers	Data	trainer_id · name · specialisation_csv · phone · contract_type · rate · bgv_status · active	Full

5.1 AppScript DAL — Mandatory Rules

- Every read and write routes through AppScript DAL functions (getStudents(), addFeeRecord(), markAttendance()) — never direct sheet.getRange().setValue()
- Every write appends to _audit_log: timestamp, user_role, action, table, row_id, old_value, new_value
- Every row has a unique auto-generated ID in Column A (PREFIX-YYYYMMDD-NNN). Never use row numbers as identifiers.
- Soft deletes only: set deleted=TRUE + deleted_at + move JSON snapshot to _deleted tab. Never call sheet.deleteRow().
- No formulas in data tabs. Raw values only. All calculations in AppScript code.
- No merged cells in any data tab.
- Continuous data from Row 2 (Row 1 = headers). No empty rows between records.
- Foreign keys enforced programmatically in AppScript before every write.
- All data tabs hidden via sheet.hideSheet() at deployment — owner sees only a blank Sheet file.
- LockService.getScriptLock() with 10-second timeout before every write to prevent concurrent access corruption.

5.2 Automated 3-Copy Backup System

Feature / Sub-module	Requirement
Copy 1 — Daily, 2:00 AM IST	DriveApp.getFileById(sheetId).makeCopy(datestamped_filename, backupFolder). Last 30 copies kept. Owner receives WhatsApp confirmation at 2:05 AM: "Backup complete — [N] records safe."

Copy 2 — Weekly, Sunday 3:00 AM IST	Export each data tab as individual CSV to MNTTA backup Google Drive account (separate Google account). Last 12 weekly exports per school. Survives school account deletion or hack.
Copy 3 — Monthly, 1st of month, 8:00 AM IST	Generate monthly summary PDF (enrolment + fees + attendance) via AppScript. Send via n8n to owner WhatsApp. Owner saves to phone as permanent offline record.
Backup health monitor	After each backup: check file created + file size reasonable + row counts consistent. If check fails: WhatsApp alert to owner AND MNTTA support.

6. Technical Architecture

6.1 V1 — Google AppScript + HTML/CSS/JS (Build This)

Feature / Sub-module	Requirement
Frontend	HTML/CSS/JavaScript served via AppScript HtmlService. Mobile-first responsive design (375px baseline). Deployed as Web App from Google Drive. Owner bookmarks the URL. Target load time: under 3 seconds on Indian 4G.
Backend / Database	Google Sheets as structured database (26 tabs per school). AppScript handles all business logic, validation, and automation. Named ranges for School Profile fields.
Authentication	Google OAuth for school owner full access. Teacher limited login: separate AppScript deployment with role-based access (attendance + observations only, no fee/CRM/management access).
Automation	AppScript time triggers: fee reminders (plan-due-date-aware) · attendance alerts (9:30 AM) · birthday check (7:00 AM) · document expiry alerts (30 days) · backup (2:00 AM). n8n webhooks: GoHighLevel sync · WhatsApp API · Google Calendar.
Infrastructure cost	₹0/month for school data (Google Drive free tier: 15GB covers 500+ schools). MNTTA infrastructure: ₹0/month for V1.
Scalability limit	AppScript 6-min execution limit · Sheets 10M cell limit · LockService for concurrent access (max 5–10 simultaneous users safely). V2 migration triggered when school exceeds 300 active students or 5 concurrent users.

6.2 V2 — React + Supabase (Build After 50 Paying Customers)

Feature / Sub-module	Requirement
Frontend	React.js + Tailwind CSS · Progressive Web App (PWA) · Installable on Android/iOS without App Store · Offline-capable for attendance and fee entry
Backend	Supabase (PostgreSQL + Auth + Storage + Realtime) · Row-level security for multi-tenancy · API-first design · Each V1 Sheet tab maps directly to a Supabase table — zero data transformation on migration
Mobile app (V3)	React Native · Android first (85%+ of Indian school owners use Android) · Biometric login

7. GUI Design System & UX Requirements

7.1 Core Design Principles

Feature / Sub-module	Requirement
One task per screen	Each screen solves one problem. One primary action button maximum. No cross-module clutter.
Three-tap rule	Any core task completable in 3 taps or fewer. If more, redesign the flow.
Default to the most common action	Attendance opens with All Present. Lead form opens with Quick Add 4-field version. Observation opens Quick 15-second entry.
Zero jargon	No "CRM", no "pipeline", no "module". Labels are real actions: "Today's Attendance", "WhatsApp to Parents", "Admissions".
Mobile-first	375px width baseline. Every screen designed for one-handed phone use. Desktop is secondary.
Forgiving interface	All deletes are soft-delete (recoverable). Autosave every 30 seconds. Confirmation required for any destructive action.
WhatsApp-native	Every "Send to Parent" action defaults to WhatsApp. Email is secondary. Always preview message before sending.
Indian cultural context	Warm colours (saffron, teal, marigold). Indian children illustrations. DD/MM/YYYY date format. ₹ currency everywhere. Indian name fields support Devanagari/Kannada/Tamil/Telugu.

7.2 Navigation — Bottom Bar (5 Tabs)

- Home — daily dashboard (attendance today, fees this week, next event, birthday alert, social media prompt, follow-up nudges)
- School — operational modules: Students · Classrooms · Attendance · Fees · Staff · Transport · Safety
- Parents — communications: WhatsApp · Photos · Circulars · PTM · Handbook · eConsents
- Marketing — social media engine · content calendar · admissions CRM · ECA promotions
- Reports — analytics dashboard · fee · attendance · ECA · annual report

7.3 Homepage — Daily Command Centre

Feature / Sub-module	Requirement
TODAY card	Live attendance % with one-tap to mark · fees collected this week · next event with date · posts published this month vs target-12
BIRTHDAY ALERT (conditional)	Shown ONLY if any child has birthday today or tomorrow. Large warm card with child photo, name, classroom, age turning. One-tap: Send Birthday WhatsApp to Parent. One-tap: Generate Birthday Social Media Post.
POST TODAY card	Today's suggested social media post from content calendar · caption preview · one-tap copy · link to Canva template

QUICK ADD button	Floating "+" button always visible. One tap → 4-field lead capture form → save. Always accessible regardless of which screen is open.
FOLLOW UP card	Leads not contacted in 48 hours · students with overdue fees · documents expiring in 30 days

7.4 Colour System

Feature / Sub-module	Requirement
Primary (action buttons, active nav)	MNTTA Teal — #148F77
Secondary (page titles, data labels)	MNTTA Navy — #1A3A5C
Accent (alerts, badges, highlights)	Warm Gold — #B7950B
Background	Warm White — #FAFAFA (not pure white — too harsh on mobile in Indian sunlight)
Text primary	Dark Charcoal — #2C3E50
Text secondary	Medium Grey — #7F8C8D
Success states (paid, present, published)	Soft Green fill #D5F5E3, text #1E8449
Warning states (due, expiring, pending)	Soft Yellow fill #FEF9E7, text #B7950B
Danger states (absent, overdue, expired)	Soft Red fill #FADBD8, text #922B21
Typography	Noto Sans (supports Devanagari/Kannada/Tamil/Telugu) · Font sizes: 28px page title · 20px section title · 16px body · 13px label · 11px hint

7.5 Onboarding — 5-Minute Setup Flow

1. School name + city + phone + WhatsApp → fills immediately across all screens (30 sec)
2. Select school type: Montessori or Kindergarten → UI adapts instantly (20 sec)
3. Upload school logo → appears immediately in Parent Handbook preview (30 sec)
4. Enter batch name + start date + max seats → Admissions CRM ready (30 sec)
5. Preview first auto-personalised social media post → copy it → first post done (60 sec)
6. Add 3 students → appear simultaneously in attendance, fees, and profiles (60 sec)

After onboarding: personalised "Your school is ready" screen with 3 suggested first actions.

8. Build Sequence & Phase Plan

Phase	Timeline	Deliverables	Revenue Target
Phase 1 — MVP	Weeks 1–4	M1 School Profile GUI · M3 Admissions CRM (with Quick Add) · M8 Social Media Engine · Deploy · Launch Early Bird ₹7,990	₹3.99L (50 buyers)
Phase 2 — Ops Core	Weeks 5–10	M2 Student Profiles · M4 Attendance (health log + pickup log + substitute) · M5 Fee Management (installment plans) · M6 Communication + Photo Sharing · M18 Classroom & Teacher Management	₹12L (100 buyers)
Phase 3 — Full Suite	Weeks 11–18	M7 PTM · M9 Staff · M10 Curriculum (quick obs) · M11 Calendar GUI + Birthday mgmt · M12 Transport · M13 Safety · M15 Handbook · M19 ECA Revenue Engine	₹22L (200 buyers)
Phase 4 — Intelligence	Weeks 19–24	M14 Reports Dashboard · M16 CPD Log · M17 n8n + GoHighLevel automations · Multi-branch Enterprise tier · Birthday social media automation	₹34L (250+ buyers)
Phase 5 — V2	Months 7–12	React + Supabase rebuild · PWA mobile install · Android app · SEA market entry (Thailand, Malaysia)	₹60L+ (400+ buyers)

9. Data Protection, Integrity & Backup

9.1 Five-Layer Protection Model

Feature / Sub-module	Requirement
Layer 1: Sheet protection	Every tab protected with password known only to AppScript service account. User editing directly → "This range is protected." End users cannot bypass. AppScript uses SpreadsheetApp bypassing protection internally.
Layer 2: DAL wall	All reads and writes through typed AppScript functions. No module calls sheet.getRange().setValue() directly. GUI never touches the Sheet.
Layer 3: Input validation	Before every write: required fields present · text max length · valid date format · numeric positive in range · FK references exist. Invalid data rejected with clear GUI error.
Layer 4: Soft deletes only	deleteRecord() sets deleted=TRUE + deleted_at + moves JSON snapshot to _deleted tab. Owner can restore from Settings → Deleted Records. Data loss physically impossible through normal app operation.
Layer 5: Audit log	Every write: append to _audit_log — timestamp, user_role, action, table, row_id, old_val, new_val. Append-only — cannot edit existing rows. Full reconstruction of every change ever made.

9.2 Recovery Scenarios

Incident	Severity	Defence	Recovery
Accidental cell edit	Low	Sheet protection blocks it	No action needed — protection worked
Row sort scrambles data	Medium	Row IDs (not row numbers) keep relationships intact	Version history restore or AppScript re-sort by ID
Record deleted	High	Soft delete to _deleted tab with JSON snapshot	Settings → Deleted Records → Restore (30 seconds)
Entire sheet deleted	Critical	Google Drive Trash retains 30 days · Copy 1 backup	Drive Trash restore — 2 minutes. Max loss: since 2AM backup
Google account hacked	Critical	Copy 2 (weekly CSV) lives in MNTTA backup account	MNTTA imports CSV to new account. Max loss: 1 week
Formula corruption in _config	Medium	AppScript nightly health check validates named ranges	AppScript auto-repairs from _config master list

10. Success Metrics

10.1 Business Metrics (MNTTA)

Feature / Sub-module	Requirement
Month 1	50 Early Bird licences · ₹3.99L revenue · 10 MNTTA graduates using system · NPS ≥ 70
Month 6	200 licences · ₹22L cumulative · 30+ schools in 5 cities · 5 Enterprise branch licences
Month 12	500 licences · ₹55L+ cumulative · 4.5/5 user satisfaction · 1 testimonial video per week
Year 2	2,000 schools · Thailand/Malaysia pilot · Android app in Play Store · 3 franchise chain partners

10.2 Product Engagement Metrics (Per School)

Feature / Sub-module	Requirement
Daily engagement	App opened ≥ 5 days/week · attendance marked by 9:45 AM · at least 1 photo shared to parents per week
Marketing engine	≥ 3 social media posts per week using templates · ≥ 40 parent enquiries/month · admission conversion ≥ 50%
Operational efficiency	Attendance marked in < 2 minutes · fee collection reconciled in < 5 minutes · WhatsApp to all parents in < 30 seconds
Financial impact	ECA programmes generating ≥ ₹50,000/month additional revenue per school within 6 months of using M19
NPS and retention	NPS ≥ 70 at 90-day mark · 40% of buyers refer at least one other school owner within 6 months

11. Appendix — Module Build Priority

Module	Business Value	Build Effort	Phase
M8 Social Media Engine	★★★★★	Low (Excel toolkit already built)	1 — MVP
M3 Admissions CRM + Quick Add	★★★★★	Medium	1 — MVP
M1 School Identity + Profile	★★★★★	Low	1 — MVP
M4 Attendance + Health + Pickup + Substitute	★★★★★	Medium	2
M5 Fee + Installment Plans	★★★★★	Medium	2
M6 Communication + Photo Sharing	★★★★★	Low-Medium	2
M18 Classroom + Teacher Management	★★★★★	Medium	2
M2 Student Profiles + Medical + Medication log	★★★★	Medium	2
M7 PTM Manager	★★★★	Medium	3
M15 Parent Handbook PDF	★★★★	Medium	3
M11 Calendar GUI + Birthday Management	★★★★	Medium	3
M19 ECA Revenue Engine	★★★★	Medium-High	3
M12 Transport	★★★★	Medium	3
M13 Safety & POCSO	★★★★	Low	3
M9 Staff + Substitute assignment	★★★	Medium	3
M10 Curriculum + Quick Observation	★★★	High (Montessori progression data)	3
M14 Reports Dashboard	★★★	Medium	4
M17 Integrations (n8n/GHL)	★★★★	High	4
M16 CPD Log (simplified)	★★	Low	4

Competitive Positioning — One Sentence

"EduManager Pro is the only preschool management system in India built by educators for educators, designed for Indian school owners and Indian parents, priced to be owned not rented, and the only system in the world that also runs your school's marketing — all for ₹14,990, once, forever."